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TITLE PAGE

- Team Name: ACE
- Theme: AI & MACHINE LEARNING
- PS Category: Software
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DEXATHON

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PROBLEM STATEMENT

- AI fails when data changes ,every fix needs proof.
- Finding the failure doesn't reveal the cause.
- A fix can break something else.
- Every fix needs proof before we trust it.
- Retraining on new data can fix one failure but quietly creat another.
- Without knowing the true cause, teams waste time and compute on fixes that don't work.

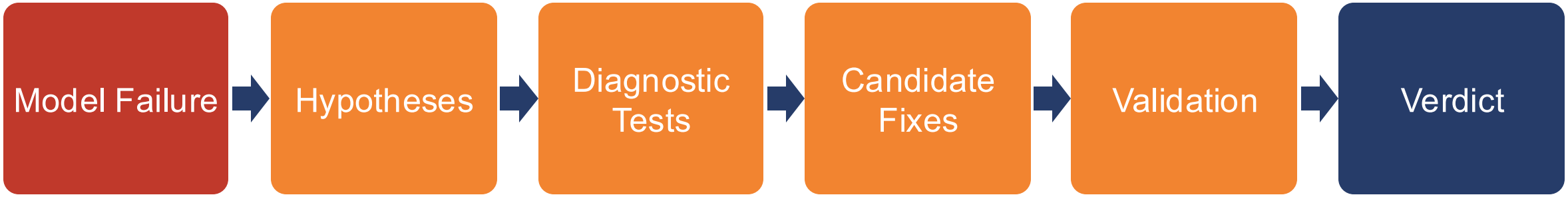
"The real challenge isn't detecting failures; it's proving that the fix actually works."

OUR SOLUTION

- Detect model failures.
- Generate competing explanations.
- Challenge explanations through targeted tests.
- Evaluate possible corrections.
- Approve a fix or abstain if evidence is insufficient.
- Check each fix on held-out data before accepting it.
- Explain every verdict with the evidence behind it.

"No evidence, no fix: uncertain solution are rejected."
"Every correction must be tested before it is trusted."

HOW IT WORKS



The original model remains unchanged until a correction passes predefined checks.

"No fix reaches the verdict without passing validation."
"The model is changed only after the evidence says the fix works."

AI / ML TECHNOLOGY

- Python, NumPy and pandas.
- Scikit-learn for model training and evaluation.
- Statistical drift detection.
- Hypothesis testing and information-gain-based test selection.
- Streamlit for the interactive interface.
- Cross-validation and a held-out test set to check whether a fix really works.
- Matplotlib/ploty charts to show drift and before-vs-after results.

These are proposed technologies; the implementation is yet to be built.

INNOVATION AND UNIQUENESS

- Combines failure investigation with evidence-based correction testing.
- Selects informative tests within a limited budget.
- Allows the system to reject uncertain solutions.
- Tests whether this strategy outperforms simpler approaches.

Simple approach : Blind retraining / random files

Budget-limited testing : picks only the most information tests

Our approch : investigate-test fix-accept or reject

IMPORTANCE AND IMPACT

- Gives clear visibility into why AI models fall and what caused it .
- Reduces fixes that don’t actually work .
- Makes model updates safer , because every fix is tested .
- Could support applications in finance, cybersecurity, healthcare research and industrial monitoring.
- Save time and compute by testing only the most informative fixes.
- Builds trust in AI systems by rejecting fixes the evidence doesn't support.

These are potential benefits to evaluate, not guaranteed outcomes.

EXPECTED OUTCOME AND FUTURE SCOPE

- A working ML failure-investigation prototype.
- A comparison of different diagnostic strategies.
- Measured repair success, false approvals and compute cost.
- Future scope: larger datasets, more model types and controlled deployment.

“An AI should not merely propose a fix. It should provide evidence that the fix deserves to be trusted.”